Case Study 1: Attendance Percentage Calculator

1. Problem Statement

Calculate a student's attendance percentage using the number of classes attended and the total number of classes.

Display whether the student is eligible based on a 75% attendance requirement.

2. Objectives

1. Read student name and attendance details.
2. Calculate attendance percentage.
3. Use an if-else condition to check eligibility.
4. Display a clear attendance report.

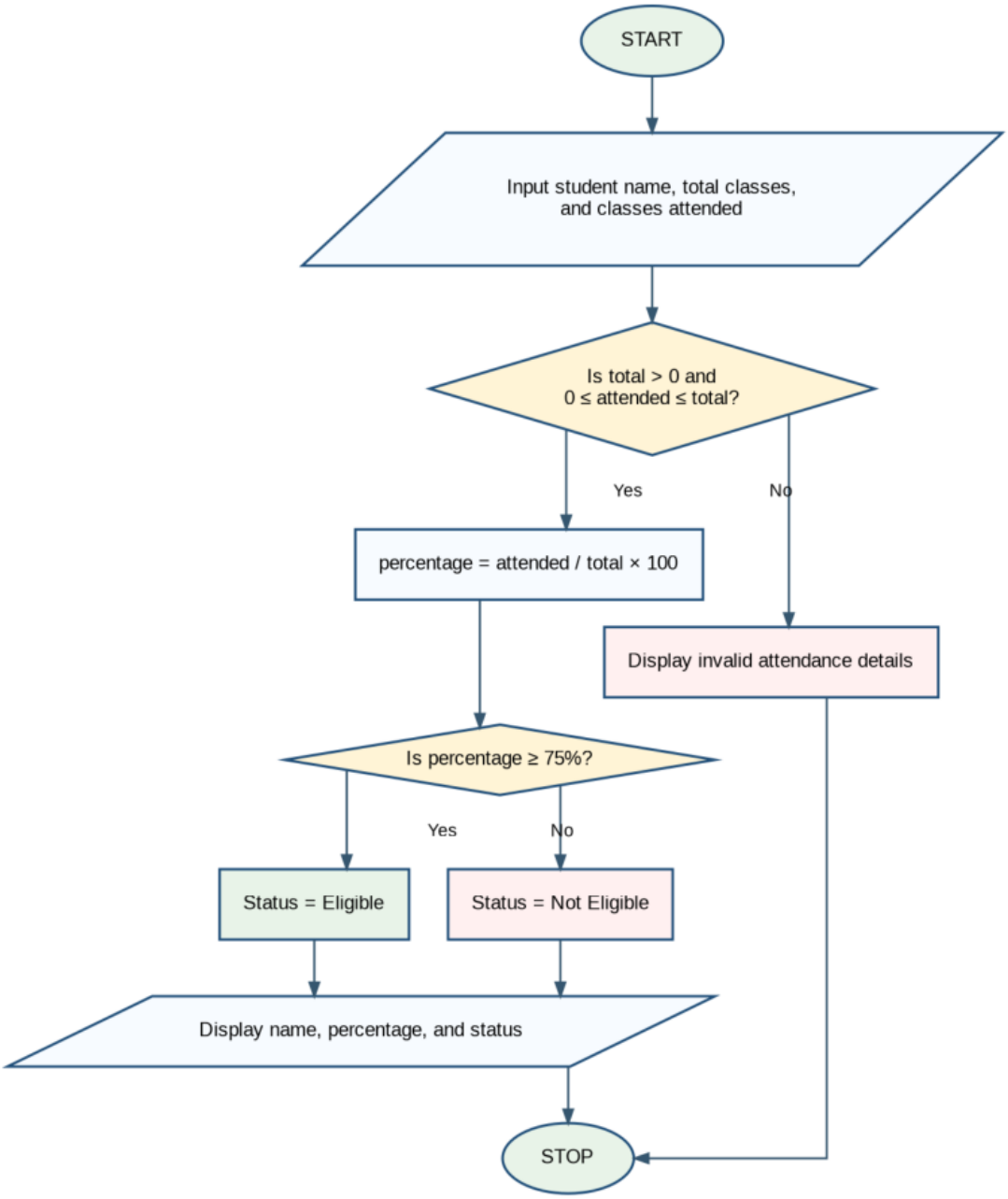
3. Assumptions

1. Total classes must be greater than zero.
2. Classes attended cannot be negative or greater than total classes.
3. Minimum required attendance is 75%.

4. Algorithm

1. Start.
2. Read student name, total classes, and classes attended.
3. Calculate percentage = (attended / total) × 100.
4. If percentage is at least 75, set status to Eligible; otherwise set status to Not Eligible.
5. Display the attendance report.
6. Stop.

5. Flowchart



6. Pseudocode

```
INPUT name, total_classes, attended
percentage = attended / total_classes * 100
IF percentage >= 75 THEN
    status = "Eligible"
ELSE
    status = "Not Eligible"
DISPLAY name, percentage, status
```

7. Complete Python Source Code

```
name = input("Enter student name: ")
total = int(input("Enter total classes: "))
attended = int(input("Enter classes attended: "))
if total <= 0 or attended < 0 or attended > total:
    print("Invalid attendance details")
else:
    percentage = attended / total * 100
    if percentage >= 75:
```

```
        status = "Eligible"
    else:
        status = "Not Eligible"
    print("\n----- ATTENDANCE REPORT -----")
    print("Student:", name)
    print(f"Attendance: {percentage:.2f}%")
    print("Status:", status)
```

8. Test Case and Expected Output

Name = Ravi; Total classes = 100; Attended = 82.

Percentage = (82 / 100) × 100 = 82%.

Expected status = Eligible.

9. Conclusion

The program calculates attendance percentage and checks eligibility using a conditional statement.